DATA CLEANING

The initial dataset comprised 53,005 movies, of which only 5,375 had non-missing budget values. During data cleaning, 10,239 movies (19.3%) were excluded due to missing values in one or more features. Notably, 9,895 movies lacked a movie_actor_score, primarily because actors with fewer than three movie appearances were excluded from the actor score calculation. This filter disproportionately affected documentaries and small-scale productions with limited actor data. The remaining missing features were attributed to movies with minimal available data, such as those in genres like documentaries, which often lack certain attributes (e.g., actor credits), or obscure films with incomplete records.

OUTLIER ANALYSIS

For the resulting 42,766 movies, a comprehensive outlier analysis was conducted to validate the data prior to machine learning training. Summary statistics for the features are presented in Table 1. Analysis of the distributions of movie_actor_score, director_quality, and movie_writer_quality revealed that fewer than 1% of movies were outliers. These outliers represented natural extremes, such as A-list actors exceeding the upper IQR bound for movie_actor_score or inexperienced writers and directors falling below the lower bound. As these outliers accurately reflected a small but significant subset of the dataset and were critical for robust model training, they were retained.

For genre_score, 6,015 movies (14.1%) were identified as outliers, with 2,376 below the lower IQR bound of 5.17 and 3,639 above the upper bound of 6.86. Further examination showed that all movies below the lower bound had a genre_score of 4.9753, corresponding to the Documentary genre, while those above the upper bound had a genre_score of 7.2133, representing the Horror genre. These outliers were meaningful, capturing genre-specific trends, and were thus retained for model training.

The budget feature exhibited the highest number of outliers, with 8,938 movies (20.9%) outside the IQR bounds ($13,201,628.41–$37,861,736.56). This was largely due to the imputation of 95% of budgets as genre-specific averages (e.g., $4,388,240.25 for documentaries), which narrowed the interquartile range (IQR) and made the bounds overly sensitive. The budget distribution was heavily right-skewed, with a long tail of high-budget films. To address this, a log transformation was applied to budget, creating an approximately normal distribution for Z-score analysis. Using a threshold of three standard deviations, 2,951 movies (6.9%) were identified as Z-score outliers in the log-transformed data. Most of these outliers had the imputed budget of $4,388,240.25 for documentaries, reflecting valid genre-specific patterns. These outliers were statistically significant and retained for machine learning training to ensure a representative dataset.